BODAPATI VISHNU SAI VARDHAN

Game Developer | XR Developer | Spatial Computing Developer

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ABOUT ME

Game and XR developer pursuing an M.Eng in Game Design, Development and Innovation at Duke University. I build immersive gameplay systems, Meta Quest mixed reality tools, clinical VR rehabilitation interactions, and AI-enabled interactive products. My work combines Unity/Unreal development, spatial interaction design, C#/C++ programming, rapid prototyping, performance-aware XR implementation, and polished user-facing presentation for hackathons, research teams, and playable game projects.

EDUCATION

Duke University — **Aug 2025 - Present**

Master of Engineering, Game Design, Development and Innovation | Pratt School of Engineering

- Key coursework: Advanced Game Design, Unreal Engine Development, C++ Programming.

University of Wollongong — **Sep 2021 - Sep 2024**

Bachelor of Computer Science, Game and Mobile Development

- Key coursework: Game Engine Essentials, VR/AR Game Development, Interactive Computer Graphics, Mobile Application Development, IT Project Management, Software Development Methodologies.

AWARDS & HACKATHONS

DesignXR Hackathon 2026 - 1st Place / Grand Winner

MR Blueprint - Mixed Reality Physics Sandbox | Devpost project | Project Link

- Won Grand Winner for a Unity-based Meta Quest mixed reality sandbox that lets users spawn, draw, edit, and simulate interactive 3D physics scenes in an XR workspace.

Team USA x Google Cloud Hackathon - Winner, Honorable Mentions

Draft USA - The LA28 Momentum League | Devpost project | Project Link

- Built a Gemini-powered fan engagement app where users draft Team USA sports, Olympic-Paralympic pairings, and hometown hubs, with transparent Momentum Score logic and responsible AI framing.

DevStudio 2026 by Logitech Hackathon - Top 50 Semifinalist

MR Blueprint - selected from a hackathon with 1,323 participants | Devpost project | Project Link

- Advanced to the Top 50 semifinalist stage with a spatial computing prototype built using Unity XR, Meta Quest, and Logitech MX Ink stylus interactions.

RESEARCH & WORK EXPERIENCE

Research Assistant - Duke Intelligent Interactive Internet of Things (I3T) Lab — **Oct 2025 - Present**

VR Gameplay & Research Developer | Duke University

- Develop VR-based rehabilitation gameplay for ICU patients, prioritizing seated/fixed-position comfort, accessibility, clear feedback, and constrained patient movement scenarios.
- Implement and iterate on Unity/Meta Quest gameplay mechanics, calibration/tutorial flows, interaction feedback, and repeated-session reliability for clinical research use.
- Work on frame-rate stability, input responsiveness, and interaction smoothness to reduce motion discomfort risk in clinical VR settings.
- Collaborate with researchers and developers to translate clinical requirements into usable gameplay systems, test flows, and patient-friendly interaction design.
- Research and prototype integration paths for Gaussian Splatting and AI-generated scenes inside Lumi to support richer, more realistic therapy environments.

Freelance Research Analyst - Singapore Gaming & Graphics Startup Ecosystem — **Oct 2024 - Dec 2024**

Exelvision IT Labs LLP

- Conducted market research across gaming, graphics, robotics, 3D modelling, VR/AR development, and artificial intelligence sectors.
- Compiled industry trends, startup performance metrics, investment activity, competitor strategies, and emerging technology opportunities for client-facing reports.
- Produced data-driven insights that supported strategic decisions, business development initiatives, and sector-specific opportunity evaluation.

SELECTED PROJECTS

MR Blueprint - Mixed Reality Physics Sandbox **Jan 2026 - Apr 2026**
Unity | C# | Meta Quest 3/3S | OpenXR | XR Interaction Toolkit | Meta XR SDK | Logitech MX Ink SDK | Devpost

- Built a Meta Quest mixed reality physics sandbox where users can spawn, edit, draw, and simulate interactive 3D physics scenes inside an XR workspace.
- Implemented core C# systems for object spawning, object selection, transform manipulation, simulation controls, world-space XR UI, content drawers, inspector panels, and object highlighting.
- Developed Edit Mode and Simulate Mode workflows, allowing users to configure mass, friction, restitution, gravity, scale, rotation, and color before testing real-time physics behavior.
- Contributed to Logitech MX Ink stylus-based Draw Mode interactions, including surface drawing, spatial drawing, pressure-sensitive strokes, undo/clear tools, repositioning, and pen connection status feedback.
- Helped implement PhysicsLens and simulation reset features, including live graphing, physics behavior visualization, and snapshot/restore workflows for rapid experimentation.

Draft USA - AI Sports Draft Strategy Platform **May 2026**
Next.js 14 | TypeScript | Tailwind CSS | Gemini API | Google Cloud Run | Docker | Recharts | Devpost

- Built a responsible AI fan engagement app that turns Team USA sport discovery into a fantasy-style draft experience focused on momentum, parity, hometown breadth, and underfollowed sports.
- Integrated Gemini-powered features for scout recommendations, roster analysis, parity coaching, leaderboard explanations, Commissioner Recaps, and responsible-language checks using server-side API routes.
- Designed user-facing dashboard flows for Draft Room, Roster Dashboard, Hometown Hubs, Momentum Score breakdowns, bonus unlocks, league storytelling, and AI scout questions.
- Implemented responsible-language constraints that frame scores as fan discovery signals rather than medal predictions, betting signals, or athlete rankings.

Tower of Tricks - UI Programmer | Build & Debugging **Aug 2025 - Dec 2025**
Unreal Engine | Blueprints | UMG | Packaging | QA

- Designed and implemented menus, HUD elements, settings/pause flows, and player-facing feedback systems for a playable Unreal Engine project.
- Debugged editor-vs-packaged build issues related to level loading, map inclusion, asset references, initialization order, and gameplay stability.
- Managed packaging configuration and performed systematic QA across builds to validate gameplay, UI behavior, and deployment readiness.

Hungry Owl - Sole Developer **Aug 2025 - Dec 2025**
Playdate | Lua | Gameplay Systems | UI

- Designed and developed a complete Playdate game from scratch, owning gameplay, systems, UI, scoring, enemy behavior, difficulty progression, and game state logic.
- Integrated hardware-specific input using the Playdate crank and buttons, mapping physical controls to responsive in-game aiming and interaction systems.
- Iteratively tested, debugged, and polished gameplay to ensure stability, responsiveness, and player clarity on constrained hardware.

Overpriced - Game Jam **Nov 2025**
Gameplay Programmer | Audio Programmer | Unreal Engine

- Implemented the core audio integration pipeline, including sound cue setup, event triggering, and contextual audio feedback tied to gameplay interactions.
- Programmed gameplay interaction logic supporting player-environment connectivity, puzzle flow, level progression, and responsive feedback states.

The Misadventures of Skylar Knight - Game Jam **Jan 2026**
Gameplay Programmer | Audio Engineer | UI Developer | Unreal Engine

- Implemented Unreal Engine Blueprint gameplay mechanics, interactions, core logic, UI/audio feedback, and 3D-to-2D gameplay adaptation.
- Conducted build testing and packaging, resolved submission-blocking issues, and supported version control workflows for team collaboration.

Meteor Mayhem - Hand Gesture-Based Game
Python | Computer Vision | Unity | C#

- Developed a gesture-controlled Unity game using Python-based real-time hand gesture recognition to drive in-game actions and natural interaction flow.

Additional Unity Academic Prototypes
FPS Prototype | VR Maze | Interactable VR Room

- Built Unity prototypes covering first-person mechanics, enemy/weapon systems, VR locomotion, spatial audio, haptic feedback, object interaction, and physics-based manipulation.

TECHNICAL SKILLS

Engines & XR: Unity, Unreal Engine, Meta Quest 3/3S, XR Interaction Toolkit, OpenXR, Meta XR SDK, Logitech MX Ink SDK, URP, world-space XR UI, spatial interaction design.

Programming: C#, C++, C, Python, JavaScript, TypeScript, Java, Lua, SQL/MySQL.

Gameplay & Systems: Gameplay programming, physics interactions, UI/HUD systems, audio implementation, game states, input systems, build packaging, QA/debugging, performance optimization.

Web, Cloud & AI: React, Next.js, Tailwind CSS, Gemini API, Google Cloud Run, Docker, server-side API routes, prompt/agent constraints, responsible AI design.

Tools: Git/GitHub, Android Studio, VS Code, Netlify, Formspree, documentation, technical research, prototyping, demo polish.

RESEARCH & DEVELOPMENT INTERESTS

- Spatial computing and mixed reality authoring tools for Meta Quest and future XR platforms.
- Clinical VR interaction design for seated, constrained, or accessibility-sensitive users.
- Gaussian Splatting, AI-generated scenes, and real-world environment reconstruction for immersive applications.
- AI-assisted interactive products that combine strong product framing, responsible generation, and polished user experience.

CERTIFICATIONS & LEADERSHIP

Certifications: Young AI Pro Training Program - Advanced AI & LLM Techniques (2024); Game Development With Unity (2020, MAGES Institute of Excellence); Professional Remote Pilot License Program (2019, Ace Aviation); UI Design Course (2020, Udemy); User Experience (UX): The Ultimate Guide to Usability and UX (2020, Udemy).

Leadership: Entrepreneur Society Captain - led fundraising and supported event organization (2018).

Languages: English (fluent), Telugu, Hindi.

Updated CV - July 2026